

Alisson Davi Gonzaga Chagas

Full-Stack Software & AI Systems Engineer

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PROFESSIONAL SUMMARY

I'm Full-Stack Software & AI Systems Engineer with hands-on experience designing and shipping production-grade applications across diverse domains, from fraud detection and automation platforms to media applications. I've specialized in building end-to-end systems with Next.js, React, TypeScript, Python, FastAPI, PostgreSQL, Redis, and Docker, while also developing and orchestrating AI Agents using LangChain, LangGraph, CrewAI, GraphiFi, RAG, embeddings, and vector databases. I'm comfortable owning the entire development lifecycle, from architecture and database design to AI integration, deployment, and production delivery.

TECHNICAL SKILLS

Languages & Frontend: Python, TypeScript/JavaScript, SQL · React, Next.js (App Router), React native, Tailwind CSS
Backend & Data: FastAPI, SQLAlchemy/Alembic, PostgreSQL, Redis, PgVector, WebSocket, JWT/OAuth 2.0 via authlib, REST, ELK Stack
AI & Architecture: LangChain, LangGraph, GraphiFi, CrewAI, RAG pipelines · Clean Architecture, SOLID, Event-Driven Design, Spec-Driven Design, Context Engineering
DevOps & Cloud: Docker, AWS, Cloudflare, Vitest, Jest, Pytest, GitHub Actions (CI/CD), Sentry, Prometheus, Grafana, OpenTelemetry

FEATURED PROJECTS

Almaginator - AI-Powered Image Generation & Editing Platform (2026)

Next.js · FastAPI · CrewAI/LangChain/LangGraph · GraphiFi · PostgreSQL · Redis · RabbitMQ · Celery · Python

- Developed a full-stack AI platform that generates and edits images from natural-language prompts, orchestrating an agentic pipeline with CrewAI/LangChain/LangGraph behind an application port (hexagonal architecture) so the framework is swappable without touching business logic;
- Engineered a LangGraph state-machine pipeline (prompt interpretation → safety review → generation/edit → post-processing) with specialized CrewAI agents, including a SafetyReviewerAgent and structured prompt delimiters as defense-in-depth against prompt injection; drove quality through Eval-Driven Development (EDD) for the AI agents alongside TDD with pytest; and I also designed asynchronous, non-blocking generation via Celery (RabbitMQ) with real-time progress streaming through Redis Pub/Sub → Server-Sent Events (SSE), plus PostgreSQL/SQLAlchemy async persistence and Redis stale-while-revalidate caching.

PaySentinelIQ - AI-Powered Payment Intelligence SaaS (2026)

Next.js · FastAPI · PostgreSQL · Redis · LangChain/CrewAI/LangGraph · PGVector + Embedding (BAAI/bge-m3) for RAG

- Built a full-stack fraud-detection platform with a 7-stage multi-agent LLM pipeline producing a 5-tier risk score (0–100) for Brazilian financial documents;
- Improved document-analysis accuracy with a RAG architecture (BAAI/bge-m3 embeddings + PgVector), reducing interpretation inconsistencies;
- Shipped CI/CD (GitHub Actions), production observability (Sentry, Prometheus, Grafana), and LGPD/DPIA compliance documentation;
- Resolved critical NextAuth authentication and App Router middleware session-persistence issues, and refactored deployment configuration across Railway, Vercel, and Cloudflare.

Pytomatiza+ - Intelligent Automation Platform (2026)

Next.js · FastAPI · CrewAI/LangChain/LangGraph · PostgreSQL · Multi-Tenant APIs

- Built a multi-domain automation platform orchestrating configurable CrewAI/LangChain agent workflows through a unified UI for non-technical users;
- Resolved production deployment issues (typing, auth endpoints, health checks) across Railway and Docker environments;
- Implemented Docker containerization for consistent local-to-cloud parity, enabling scalable and reproducible Railway deployments.

ClarkPlayer - Full-Stack Music Player Application (2025–2026)

Next.js · FastAPI · Redis · Zustand · Resend

- Engineered Redis-backed session management and authentication flows, plus virtualized rendering (react-window) for large media libraries;
- Integrated MusicBrainz, Spotify, iTunes, and Genius APIs into a unified metadata-enrichment layer;
- Built drag-and-drop playlist management (dnd-kit) and a karaoke lyrics mode, resolving complex sidebar/player-bar layout conflicts via a shared Zustand store.

FacilitBot - WhatsApp Customer Service Automation (2025)

Flask · CrewAI/LangChain · Z-API (WhatsApp) · React Native (Expo) · Python

- Developed an AI-powered WhatsApp automation bot for a home-appliance business, handling customer-service conversations via CrewAI agents and the Z-API integration;
- Completely migrated the frontend from Kivy to React Native (Expo), resolving npm peer-dependency conflicts through a full scaffold rebuild;
- Restructured frontend–backend–WhatsApp communication for automated service stability;
- Refactored agent flow to produce more organized and consistent conversation responses.

EDUCATION & CERTIFICATIONS

BSc, Software Engineering - UNICESUMAR (in progress, 5th semester) · Technical Degree, Information Technology - ICEIA (2023) · Self-taught full-stack engineer since 2022 (FastAPI, Next.js, PostgreSQL, Docker, AWS, CrewAI/LangChain)
Certifications (Udemy, 334 combined hours): JavaScript & TypeScript · Python 3 · Git & GitHub · Advanced Excel

LANGUAGES

Portuguese - Native · English - Professional Working Proficiency (CEFR B2) | TOEFL ITP: 553/677 (August 2026) | CCAA · (English Language Course - Beginner to Master Level · Completed 11 semesters)